

CODING & SOLUTION

EDUGENIE – AI-POWERED LEARNING ASSISTANT

EduGenie is implemented as a Python-based FastAPI web application with separate AI-learning modules and an HTML/CSS frontend.

IMPLEMENTATION COMPONENTS

`main.py`:

Main FastAPI application and API routes.

`gemini_client.py`:

Gemini configuration and communication.

`qna.py`:

Question-answering logic.

`explanation_module.py`:

Local concept explanation using LaMini-Flan-T5-783M.

`quiz_module.py`:

Multiple-choice quiz generation.

`summary_module.py`:

Text summarization.

`learning_path.py`:

Structured learning-path generation.

`templates/index.html`:

Frontend page structure.

`static/style.css`:

Frontend styling.

`requirements.txt`:

Python dependency list.

SECURITY

The Gemini API key is supplied through an environment variable. The real `.env` file should remain local and should not be committed to a public repository.